Boundary-free weak influences: range closure, sharp information and feasible GLS

1. Spaces and adjoint range

Let $J \in C^1([0,T];\mathbb{R}^{d\times d})$, and let the invertible fundamental matrix satisfy $Z' = JZ$, $Z(0) = I_d$. Put $\mathcal{D}v = v' - Jv$ and $\mathcal{D}^*\phi = -\phi' - J^T\phi$. Sensor coordinates have independent stationary OU covariance with fixed positive scales. Its Cameron–Martin space $\mathcal{H}$ is H^1 with the OU inner product. Write $\mathcal{E} = \mathcal{H}'$ for its continuous dual, equipped with the covariance norm

$$\|\lambda\|_C^2 = \langle \lambda, C\lambda\rangle, \qquad \langle C^{-1}g, h\rangle = \langle g, h\rangle_{\mathcal{H}}.$$

This definition includes ordinary integrated influences and endpoint evaluation measures. General dual influences act through the $L^2(P)$ closure of integrated Gaussian observations, rather than a pathwise pairing for every distribution. Their variance norm is $\|\lambda\|_C$; the noise path itself need not belong to $\mathcal{H}$.

Proposition 2 (exact range and closure). In L^2,

$$\mathcal{D}^* H_0^1 = \left\{ \ell \in L^2 : \int_0^T Z(t)^T \ell(t) dt = 0 \right\}.$$

The range of compactly supported smooth tests is dense in this space in L^2. Its closure in $\mathcal{E}$ is exactly

$$\mathcal{M} = \overline{\mathcal{D}^* C_c^\infty}^{\|\cdot\|_C} = \{\lambda \in \mathcal{E} : \langle \lambda, Z_j\rangle = 0, j = 1, \dots, d\}.$$

Consequently $\mathcal{M}$ has codimension d.

Proof. Since $Z^T\mathcal{D}^*\phi = -(Z^T\phi)'$, zero endpoint values imply the moment constraint. Conversely, for an L^2 influence with zero moment,

$$\phi(t) = -Z(t)^{-T} \int_0^t Z(s)^T \ell(s) ds$$

belongs to H_0^1 and satisfies $\mathcal{D}^*\phi = \ell$. Approximation in H_0^1 by C_c^∞ and continuity of $\mathcal{D}^* : H_0^1 \to L^2$ give L^2 density. For the stronger closure statement, a function $v \in \mathcal{H}$ annihilates every $\mathcal{D}^*\phi$ if and only if $\mathcal{D}v = 0$ distributionally, hence $v = Zc$. The Hahn–Banach annihilator characterization of

a closed subspace then gives the stated closure in $\mathcal{H}'$. The columns of Z are linearly independent elements of $\mathcal{H}$, so their dual constraints have rank d. □

The exact range uses zero-endpoint Sobolev tests; literal compact tests require closure. The covariance norm is equivalent to the dual Sobolev norm for this positive OU model.

Lemma 1 (a concrete dense test family). Let V_m be vector-valued continuous piecewise-linear splines on the nested uniform dyadic partitions of $[0,T]$, with zero values at both endpoints. Then

$$\overline{\bigcup_m \mathcal{D}^* V_m}^{\|\cdot\|_C} = \mathcal{M}.$$

Proof. Given $v \in H_0^1$, choose $\psi \in C_c^\infty$ close to v in H^1. Its nodal interpolant $I_m\psi \in V_m$ satisfies $\|I_m\psi - \psi\|_{H^1} \le c\Delta_m \|\psi''\|_{L^2}$, where Δ_m is the mesh width. The union of V_m is therefore dense in H_0^1. Continuity of $\mathcal{D}^* : H_0^1 \to L^2$ and of the embedding $L^2 \hookrightarrow \mathcal{H}'$ transfers this approximation to the covariance norm. Proposition 2 identifies the closure of $\mathcal{D}^* H_0^1$ with $\mathcal{M}$. □

These splines are admissible zero-endpoint Sobolev tests; they need not vanish in an endpoint neighborhood. If literal smooth compact tests are required, approximate each spline in H_0^1 by such tests. Thus Lemma 1 provides an explicit family for the density condition in Corollary 1. It does not assert density of the finite multiscale bump families used in the experiments.

2. Sharp covariance reference

Let $S = [S_1, \dots, S_p] \in \mathcal{H}^p$ be coefficient sensitivities, and form the joint information blocks for S and Z. Assume $\mathcal{J}_{uu}$ and the Schur complement are positive definite. Set

$$R = S - Z\mathcal{J}_{uu}^{-1}\mathcal{J}_{u\theta}, \qquad \mathcal{J}_e = \mathcal{J}_{\theta\theta} - \mathcal{J}_{\theta u}\mathcal{J}_{uu}^{-1}\mathcal{J}_{u\theta}.$$

Projection and best linear unbiased estimation in correlated continuous regression are classical [1]. Here the adjoint-range characterization identifies exactly which dynamical weak influences obey that projection.

Theorem 2 (sharp boundary-free influence covariance). Let $\Lambda = [\lambda_1, \dots, \lambda_p] \in \mathcal{M}^p$ satisfy $\langle \lambda_r, S_s \rangle = \delta_{rs}$. Its Gaussian influence covariance obeys

$$V(\Lambda) \succeq \mathcal{J}_e^{-1}.$$

The bound is attained in $\mathcal{M}$ by the matrix-valued dual influence

$$\Lambda_* = C^{-1} R\, \mathcal{J}_e^{-1}.$$

It is the infimum for normalized compact-test influences. It is a minimum among literal compact tests only when Λ_* belongs to their adjoint range.

Proof. The $\mathcal{H}$ projection makes $\langle C^{-1}R, Z\rangle = 0$ and $\langle C^{-1}R, S\rangle = \mathcal{J}_e$. Thus Λ_* meets both constraints and has covariance $\mathcal{J}_e^{-1}$. The difference $\Delta = \Lambda - \Lambda_*$ pairs to zero with both S and Z, hence with R. Covariance cross terms vanish and

$$V(\Lambda) = \mathcal{J}_e^{-1} + V(\Delta).$$

For attainability as an infimum, approximate each column of Λ_* by compact-test influences in covariance norm. Their normalization matrices converge to I_p by continuous pairing with S; right-multiplication by their inverse transpose enforces exact normalization while preserving the range and covariance convergence. □

For an unknown initial state, $\mathcal{J}_e$ is the full likelihood nuisance-adjusted information, so this weak restriction has no further first-order cost. When the initial state is known, the information loss is exactly $\mathcal{J}_{\theta u}\mathcal{J}_{uu}^{-1}\mathcal{J}_{u\theta}$; covariance inflation is $\mathcal{J}_e^{-1} - \mathcal{J}_{\theta\theta}^{-1}$. There is no direction-independent scalar loss percentage in general. When u is known, $Y(0) - u$ also reveals the initial stationary measurement error, which remains correlated with later observations. The unrestricted score can exploit that observation while the weak influence must retain its initial-sensitivity annihilation constraint.

3. Explicit OU representative

For a scalar coordinate with $g \in H^2$, integration by parts in the stationary OU norm gives the dual measure

$$C^{-1}g = \frac{g - \tau^2 g''}{2\sigma^2\tau}\,dt + \frac{g(0) - \tau g'(0)}{2\sigma^2}\delta_0 + \frac{g(T) + \tau g'(T)}{2\sigma^2}\delta_T.$$

For independent coordinates the formula applies coordinatewise, using their respective scales. If only $g \in H^1$, the Riesz representative remains well-defined in $\mathcal{H}'$ but need not be a density plus finite endpoint measures.

For a smooth optimal representative write $\Lambda_* = m(t)dt + m_0\delta_0 + m_T\delta_T$. On the open interval an adjoint test matrix is

$$\Phi_*(t) = -Z(t)^{-T}\left[Z(0)^T m_0 + \int_0^t Z(s)^T m(s)ds\right].$$

Extend it by zero outside the observation interval. Its endpoint jumps supply $m_0 = -\Phi_*(0+)$ and $m_T = \Phi_*(T-)$, and distributionally $\mathcal{D}^*\Phi_* = \Lambda_*$. These formulas include the initial atom; omitting it in the cumulative integral is incorrect. The annihilation condition supplies the terminal jump. Cutoff and smoothing approximate this generalized test in covariance norm. These measures describe a statistical reference; the implemented estimators keep compact tests and add no explicit boundary observation terms.

For scalar exponential dynamics, $z = e^{at}$, $s = x_0te^{at}$, $k = \mathcal{J}_{au}/\mathcal{J}_{uu}$ and $g = e^{at}(x_0t - k)$. Define

$$A_0 = \frac{(1-\tau^2a^2)x_0}{2\sigma^2\tau}, \qquad B_0 = \frac{-(1-\tau^2a^2)k - 2\tau^2ax_0}{2\sigma^2\tau},$$

$$q_0 = \frac{-k - \tau(x_0 - ak)}{2\sigma^2}, \qquad F_j(t) = \int_0^t s^j e^{2as}ds.$$

Then

$$\phi_*(t) = -\frac{e^{-at}}{\mathcal{J}_e}[q_0 + A_0 F_1(t) + B_0 F_0(t)].$$

For $a \neq 0$, $F_0 = \mathrm{expm1}(2at)/(2a)$ and $F_1 = [e^{2at}(2at-1)+1]/(4a^2)$; their continuous limits at $a = 0$ are t and $t^2/2$. A pilot coefficient substitutes into this expression before a compact cutoff is applied. The explicit multivariate fundamental-matrix expression is an analytical identity, not a guarantee of stable numerical inversion of $Z(t)$; strongly contracting dynamics can make that inversion ill-conditioned.

4. GLS and the order of limits

For tests ϕ_k, put $\lambda_k = \mathcal{D}^*\phi_k$, $D_{kr} = \langle \lambda_k, S_r \rangle$ and $\Sigma_{k\ell} = \langle \lambda_k, C\lambda_\ell \rangle$. Remove exact linear dependencies. If D has full column rank, covariance-weighted weak regression has local influence and covariance

$$P = (D^T\Sigma^{-1}D)^{-1}D^T\Sigma^{-1}, \qquad V_K = (D^T\Sigma^{-1}D)^{-1}.$$

WENDy's covariance iteration motivates this construction [2], but the released iid covariance is not the correct physical-OU covariance. Under polynomial features Proposition 3 supplies the complete coupled residual weights. Its order-one term determines the small-noise efficient influence; higher orders correct residual uncertainty at finite amplitude without by themselves removing coefficient-estimator bias.

The same leading map arises from a parameter-dependent Gaussian weak likelihood of the form studied by WENDy-MLE [3], provided the covariance shape is correct and the local solution is regular. For fixed dimension write $r(\theta) = \epsilon r_1 + o_p(\epsilon)$, $\partial_\theta r = -D + o_p(1)$ and $\Sigma_\epsilon(\theta) = \epsilon^2\Sigma_0(\theta)$, with smooth positive definite Σ_0. The objective $\log\det\Sigma_\epsilon + r^T\Sigma_\epsilon^{-1}r$ has covariance-derivative and log-determinant score terms of order $O_p(1)$ in an $O_p(\epsilon)$ neighborhood; its mean-residual score is of order $O_p(\epsilon^{-1})$. The local score equation therefore gives

$$\hat{\theta} - \theta = \epsilon(D^T\Sigma_0^{-1}D)^{-1}D^T\Sigma_0^{-1}r_1 + o_p(\epsilon).$$

Thus the log determinant and covariance derivatives distinguish finite-noise weak likelihood from frozen GLS, while both share the correct-covariance leading influence under these assumptions. This connects Corollary 1 to a regular local weak-likelihood estimator; it does not supply global optimization or covariance-calibration consistency for a released implementation.

Corollary 1 (small-noise, dense-span efficiency). Let the noise be $\epsilon\eta_C$ with known covariance shape C. Assume a smooth identifiable parameterization, a generating parameter interior to a compact parameter set, consistent pilots, consistent covariance-shape weights and quadrature error $o(\epsilon)$. For each fixed finite test family assume a regular local weak solution with the GLS linearization. For the covariance conclusion additionally assume uniform integrability of $\|\epsilon^{-1}(\hat{\theta}_{K,\epsilon} - \theta)\|^2$. A sufficient implementation uses a uniformly stable smooth local solve, Gaussian moment control on the local Taylor remainder, and bounded fallback outside the regular region on a bad event whose probability is $o(\epsilon^2)$. Then

$$\epsilon^{-1}(\hat{\theta}_{K,\epsilon} - \theta) \Rightarrow N(0, V_K), \qquad \epsilon^{-2}\,\mathrm{Cov}(\hat{\theta}_{K,\epsilon}) \to V_K.$$

If nested test-influence spans are dense in $\mathcal{M}$, then

$$V_K \longrightarrow \mathcal{I}_{e,C}^{-1}.$$

Proof. For fixed K, a first-order expansion of the weak equations gives $D(\hat{\theta}-\theta) = \epsilon r_1 + o_p(\epsilon)$ with the GLS map above. Consistent pilots and weights change the leading map by $o_p(1)$; derivatives of pilot-dependent tests multiply the zero clean residual. Gaussianity gives the distributional assertion; the stated risk condition justifies covariance convergence. In the Gaussian dual space, finite-family information is the Gram matrix of the orthogonal projection of $C^{-1}R$ onto its test span. Density gives projection convergence, and nested spans give increasing information; continuity of inversion at positive definite $\mathcal{I}_e$ gives covariance convergence. □

The conclusion is sequential: first small noise and controlled quadrature for fixed K, then dense spans. A diagonal joint limit requires uniform remainder, conditioning and covariance-weight control that are not supplied here. This theorem concerns regular WENDy-like weak estimation with correct covariance and supplied identifiable structure. The conditional result covers the leading GLS map, rather than an algorithm name or finite-noise support-selection procedure. A fixed basis cutoff must be compatible with the density assumption; discarding an informative direction permanently can invalidate it.

References

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